

Manushri Swaminathan

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PROFILE

Aspiring Software Engineer with a strong foundation in algorithms, machine learning, and systems development. Drawn to building reliable, scalable systems — with growing interest in distributed systems, AI applications, and secure software engineering.

EDUCATION

Sardar Vallabhbhai National Institute of Technology (NIT Surat) 2024 – Present
B.Tech in Computer Science and Engineering CGPA: 8.82

- Admitted via JEE Main 2024 — AIR 3807 (General), **99.9 percentile** in Mathematics

Coursework: Information Security, Computer Networks, Artificial Intelligence, Design and Analysis of Algorithms, Python Programming, DBMS, Data Structures and Algorithms, Object-Oriented Programming

PROJECTS

JEE College Predictor [Git Repo](#) Jun 2026 - Present
React, Node.js, Express.js, PostgreSQL, Python (ML)

- Built a full-stack ML system predicting JEE counselling outcomes using **6 years (2019–2025) of JoSAA/CSAB** cut-off data, processing and cleaning **~500K records** into a normalized PostgreSQL schema
- Designing a **multi-factor ranking algorithm** that re-scores eligible colleges using NIRF ranking, median placement package, and fee structure — going beyond simple rank cutoff matching offered by existing tools
- Engineering the data pipeline handling inconsistent multi-year formats, duplicate counseling rounds, and category/quota wise rank variations across 6 admission cycles
- Planning to architect REST APIs with Express.js/Node.js to serve real-time predictions to a React frontend

Smart-Sight - Assistive Navigation Device for the Visually Impaired | [Working Video](#) | [Git Repo](#) Feb 2026 - May 2026
C/C++, Arduino, HC-SR04, Water Sensor, NEO-6M GPS, SIM800L GSM

- Designed a **two-module embedded safety system** on separate microcontrollers: a sensor unit (ultrasonic + water sensor) that triggers an audio buzzer on obstacle/hazard detection, and an independent emergency unit (GPS + GSM) for caretaker alerts
- Built the emergency module by integrating a NEO-6M GPS and SIM800L GSM on a dedicated Arduino — on a button press, the device parses live GPS coordinates and sends them via SMS (AT commands) to a registered caretaker's number
- Programmed serial communication and module initialization for both GPS (NMEA sentence parsing) and GSM (AT command handling) in embedded C/C++, independent of the obstacle-detection subsystem

AWARDS & ACHIEVEMENTS

- Earned the **Developing Secure Software (LFD121)** certification from the Linux Foundation, strengthening expertise in secure coding practices [View Credential](#) Jun 2026
- Qualified Round 1 (CTF) of E-Rakshak, a cybersecurity hackathon** by Surat City Police & SVNIT; advancing to mentored project round (Aug 2026) Jun 2026
- Ranked All India 2433, ICPC Asia Chennai Preliminary Contest 2025 Oct 2025
- Solved **300+ problems** across LeetCode, CodeChef, and Codeforces
- Completed **Google Cloud Study Jams** under Google Developer Groups on Campus initiative. [View Credential](#) Oct 2025
- Completed **Exploring Networking with Cisco Packet Tracer** – Networking Academy. [View Certificate](#)

SKILLS

Programming: C, C++, Python, JavaScript
Embedded Systems: Arduino, UART/Serial Communication, Sensor Interfacing (Ultrasonic, GPS, GSM)
Machine Learning: NumPy, Pandas, Scikit-learn, TensorFlow
Web Technologies: HTML, CSS, JavaScript, React, Node.js, Express.js, PostgreSQL, MySQL
Tools & Platforms: Git, GitHub, VS Code, Linux

LANGUAGES

English, Hindi, Tamil, Gujarati (Native/Fluent)